AQUEN FLUID TECHNOLOGIES LLP

Complete Water & Wastewater Treatment Solutions

AQUEN — FLUID BY NATURE

Design | Manufacturing | Supply | Installation | Commissioning | AMC

SEWAGE TREATMENT PLANT (STP)

MBBR TECHNOLOGY

CAPACITY RANGE : 50 KLD TO 500 KLD | OPERATION HOURS : 22 HRS

- Compact & Efficient Design
- High Treatment Efficiency
- Low Sludge Generation
- Suitable for Residential, Commercial, Institutional & Industrial Applications
- Stable Performance
- Low Maintenance
- Modular / Package Options
- Provision for Tertiary Treatment, Disinfection & Reuse

CLEANER WATER GREENER TOMORROW



PROCESS FLOW DIAGRAM – STP (MBBR TECHNOLOGY)



TYPICAL INLET & OUTLET PARAMETERS (DOMESTIC SEWAGE)

Parameter	Inlet (Typical)	Outlet (Typical)
pH	6.5 – 8.5	6.5 – 8.0
BOD (mg/L)	200 – 300	< 10
COD (mg/L)	400 – 600	< 50
TSS (mg/L)	200 – 300	< 10
Oil & Grease (mg/L)	10 – 20	< 5
Ammonia – N (mg/L)	20 – 40	< 5
Total Nitrogen (mg/L)	30 – 50	< 10
Fecal Coliform (MPN/100 mL)	$10^6 – 10^8$	< 10^3

MBBR MEDIA



- High surface area
- Durable & long life
- Excellent biofilm growth
- High treatment efficiency

TECHNICAL SPECIFICATIONS – 50 KLD TO 500 KLD (OPERATION HOURS : 22)

Parameter	50 KLD	100 KLD	200 KLD	300 KLD	500 KLD
Design Flow (KLD)	50	100	200	300	500
Average Flow (m^3/hr)	2.5	5	10	15	25
Hydraulic Retention Time (hrs)	8 – 10	8 – 10	8 – 10	8 – 10	8 – 10
MBBR Media Fill (%)	40 – 50	40 – 50	40 – 50	40 – 50	40 – 50
MBBR Media Quantity (m^3)	8 – 12	16 – 24	32 – 48	48 – 72	80 – 120
Air Requirement (m^3/hr)	60 – 80	120 – 160	240 – 320	360 – 480	600 – 800
Blower Capacity (m^3/hr) (1 + 1)	80 (5 HP)	160 (7.5 HP)	320 (10 HP)	480 (15 HP)	800 (20 HP)
Connected Load (kW) (Approx.)	8 – 12	12 – 18	20 – 30	30 – 45	45 – 70
Area Required (m^2) (Approx.)	60 – 80	80 – 110	150 – 200	200 – 280	300 – 400

Note: Values are indicative and may vary based on influent quality, site conditions and detailed design.

MOC (MATERIAL OF CONSTRUCTION)

Civil Tanks	RCC / HDPE / MSEP (MS Epoxy Painted) / FRP
Package Plant	MSEP (MS Epoxy Painted) / FRP
Piping	UPVC / CPVC / GI (as per service)
Pumps	CI / SS-304 (as per duty)
Blowers	Root Blower / Turbo Blower
Diffuser	EPDM Fine Bubble Diffuser
MBBR Media	HDPE
Screens	SS-304
Tube Settler	FRP / PP
PSF & ACF Vessels	FRP / MSEP / MSRL
Instrumentation	Level Switch, Flow Mater, DO Mater, pH Meter, etc.
Control Panel	MCC with PLC (Optional)

EQUIPMENT LIST (TYPICAL BOM)



ACTUAL SITE PHOTOGRAPHS (ONGOING / COMPLETED PROJECTS)



50 KLD STP (Package - MSEP) | 100 KLD STP (Civil + E&M) | 300 KLD STP (MBBR) | 500 KLD STP (MBBR)

APPLICATIONS

Residential Societies | Commercial Buildings | Hotels & Resorts | Hospitals | Educational Institutions | Industrial Areas | IT Parks | Malls & Multiplexes

WHY AQUEN ?

- Customized Design
- Quality Equipment
- On-time Execution
- Competitive Pricing
- After Sales Support & AMC

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WTP | STP | ETP | ZLD | FABRICATION | SPARES | AMC | RETROFIT | AUTOMATION

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1. STP MBBR – 100 KLD (Civil Work in Progress)



2. STP MBBR – 100 KLD (Electro – Mechanical Installation)



3. STP MBBR – 250 KLD (Civil Work in Progress)



4. STP MBBR – 250 KLD (Electro – Mechanical Installation)



5. Package STP MBBR – 60 KLD (Installation in Progress)



6. STP MBBR – 175 KLD (Completed Installation)

